Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/15/2023
PI Name:	Stray, Stephen J. Ph.D.		
Project Title:	Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	SSTRAY@UMC.EDU	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	University of Mississippi Medical Center		
PI Address 1:	2500 North State Street		
PI Address 2:			
PI Web Page:			
City:	Jackson	**State:**	MS
Zip Code:	39216	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/01/2020	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	The previous Principal Investigator (PI) was Dr. Ritesh Tandon at the University of Mississippi Medical Center from January 2020-October 2023. Dr. Stephen J. Stray became PI on Oct 5, 2023.		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Mehta, Satish Ph.D. (NASA Johnson Space Center) Simpson, Richard Ph.D. (University of Arizona)		
Grant/Contract No.:	80NSSC19K1603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This project (grant number 80NSSC19K1603), is a continuation of the investigation titled, "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication", with Dr. Stephen J. Stray from the University of Mississippi Medical Center as the new Principal Investigator (PI). Opportunistic pathogens such as latent viruses pose a unique hazard to crew members because of the ineffectiveness of preflight quarantine periods and decreased immune function. Latent herpesviruses reactivate and shed in higher quantities during and after long and short spaceflights. Human cytomegalovirus (CMV) is an important herpesvirus from a public health viewpoint that causes severe systemic infections upon primary infection or on reactivation from latency. The few antiviral drugs available to treat CMV infection suffer from safety, efficacy, and resistance issues, and no vaccine against CMV is currently available. As NASA and collaborating space agencies plan for future exploration class missions (i.e., to Mars or an asteroid), eliminating the risk of an adverse health event due to alterations in virus latency, reactivation, and lytic replication is paramount. Evidence of impaired control of latent viral infections has been documented during short (i.e., 14-day Space Shuttle missions) as well as long (i.e., 6-month International Space Station missions) duration spaceflight missions, with the magnitude of this response being positively associated with time in flight. Shedding of herpesviruses including Epstein–Barr virus (EBV), varicella-zoster virus (VZV), and CMV was detected in >30% of astronauts in a recent study (Mehta et al., 2017) looking at long-duration spaceflight (60–180 days). These viruses reactivated independently of each other and reactivation frequency increased in duration, and amplitude (viral copy numbers) when compared to short duration (10 to 16 days) space shuttle missions. Thus, it is easy to speculate that the substantially longer durations being proposed for exploration class missions (i.e., up to 3 years) will further increase the risk of an adverse health event due to increased virus reactivation and/or replication. Add increased radiation exposure (both duration and magnitude), limited clinical care, and the impossibility of rapidly returning a casualty back to Earth to the list of potential hazards and it is clear to see why obtaining a better understanding of virus replication under space-like conditions is critical to successfully and safely send humans to, and return them from, these remote extraterrestrial destinations. In this proposal, we aim to determine the impact of space radiation on CMV latency, reactivation, and lytic replication. While most studies on this topic in the past have focused on changes in the human immune system that could induce virus reactivation, we will approach this topic from both virus and host standpoints. We will study differences in CMV latency and reactivation before and after radiation exposure and determine if these changes can be correlated with specific changes in virus genome and/or transcriptome. Also, because virus control, dissemination, and resolution are ultimately governed by the competency, or lack thereof, of the host immune system, we will determine the impact of pre-exposing NK-cells and T-cells to space radiation on their ability to control active CMV infections. In this project, we will focus on a single variable: space radiation, rather than looking at multiple variables such as microgravity, physical and psychological stresses of space travel, diet, medications, or other factors associated with spaceflight. The overarching hypothesis is that modeled space radiation will induce specific changes in CMV genome/transcriptome or immune control of infection that will induce CMV reactivation and/or increase in virus replication and pathogenicity. The specific aims to test this hypothesis are: Specific Aim 1: Determine the impact of space radiation on CMV genome and transcriptome. Specific Aim 2: Correlate radiation induced changes in CMV genome and/or transcriptome with CMV latency, reactivation, and lytic replication. Specific Aim 3: Determine the impact of space radiation on immune cell control in a mouse model of CMV infection. Reference: Mehta SK, Laudenslager ML, Stowe RP, Crucian BE, Feiveson AH, Sams CF, Pierson DL. (2017) Latent virus reactivation in astronauts on the international space station, NPJ Microgravity 3, 11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	NOTE: This project (grant number 80NSSC19K1603), is a continuation of the investigation titled, "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication", with Dr. Stephen J. Stray from the University of Mississippi Medical Center as the new Principal Investigator (PI). For previous reporting, see "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication" (PI: Tandon).
Bibliography Type:	Description: (Last Updated: 12/18/2024)
Articles in Peer-reviewed Journals	Dwivedi R, Farrag M, Sharma P, Shi D, Shami AA, Misra SK, Ray P, Shukla J, Zhang F, Linhardt RJ, Sharp JS, Tandon R, Pomin VH. "The sea cucumber Thyonella gemmata contains a low anticoagulant sulfated fucan with high anti-SARS-CoV-2 actions against wild-type and delta variants." J Nat Prod. 2023 Jun 23;86(6):1463-75. https://doi.org/10.1021/acs.jnatprod.3c00151 ; PMID: 37306476; PMCID: PMC10401483 , Jun-2023
Articles in Peer-reviewed Journals	Farrag M, Dwivedi R, Sharma P, Kumar D, Tandon R, Pomin VH. "Structural requirements of Holothuria floridana fucosylated chondroitin sulfate oligosaccharides in anti-SARS-CoV-2 and anticoagulant activities." PLoS ONE. 2023 May 11;18(5):e0285539. https://doi.org/10.1371/journal.pone.0285539 ; PMID: 37167245; PMCID: PMC10174540 , May-2023
Articles in Peer-reviewed Journals	Maurya AK, Sharma P, Samanta P, Shami AA, Misra SK, Zhang F, Thara R, Kumar D, Shi D, Linhardt RJ, Sharp JS, Doerksen RJ, Tandon R, Pomin VH. "Structure, anti-SARS-CoV-2, and anticoagulant effects of two sulfated galactans from the red alga Botryocladia occidentalis." Int J Biol Macromol. 2023 May 31;124168. https://doi.org/10.1016/j.ijbiomac.2023.124168 ; PMID: 36963552; PMCID: PMC10175164 , May-2023
Articles in Peer-reviewed Journals	Pramanik A, Mayer J, Sinha SS, Sharma PC, Patibandla S, Gao Y, Corby LR, Bates JT, Bierdeman MA, Tandon R, Seshadri R, Ray PC. "Human ACE2 peptide-attached plasmonic-magnetic heterostructure for magnetic separation, surface enhanced Raman spectroscopy identification, and inhibition of different variants of SARS-CoV-2 infections." ACS Appl Bio Mater. 2022 Sep 2. https://doi.org/10.1021/acsabm.2c00573 ; PMID: 36053723 , Sep-2022

Articles in Peer-reviewed Journals	Pramanik A, Sharma PC, Patibandla S, Gao Y, Ruppa-Kasani V, Goli J, Kumar A, Chatterjee A, Sinha SS, Bates JT, Bierdeman MA, Tandon R, Ray PC. "Blocking SARS-CoV-2 Delta variant (B.1.617.2) spike protein receptor-binding domain binding with the ACE2 receptor of the host cell and inhibiting virus infections using human host defense peptide-conjugated graphene quantum dots." ACS Omega. 2022 Mar 8;7(9):8150-7. https://doi.org/10.1021/acsomega.2c00113 ; PMID: 35252734; PMCID: PMC8886715 , Mar-2022
Articles in Peer-reviewed Journals	Harris HM, Boyet KL, Liu H, Dwivedi R, Ashpole NM, Tandon R, Bidwell GL 3rd, Cheng Z, Fassero LA, Yu CS, Pomin VH, Mitra D, Harrison KA, Dahl E, Gurley BJ, Kotha AK, Chougule MB, Sharp JS. "Safety and pharmacokinetics of intranasally administered heparin." Pharm Res. 2022 Mar 2;1-11. https://doi.org/10.1007/s11095-022-03191-4 ; PMID: 35237922; PMCID: PMC8890767 , Mar-2022